

Name:		 MANIPAL UNIVERSITY JAIPUR	
Enrolment No:			
Summer/Makeup Examination, July- 2025 Faculty of Science, Technology and Architecture School of Computer Science and Engineering Department of Data Science B.Tech. Data Science			
Course Code: DS4145		Course: Social Network Analysis	Semester: VII
Time: 03 hrs.		Max. Marks: 80	
Instructions: All questions are compulsory. Missing data, if any, may be assumed suitably. Calculator is allowed.			
SECTION A (Memory Based Questions)			
S.No.		Marks	CO
Q 1	What does the term social web refer to? a) Websites used only for business purposes b) Platforms enabling social interaction and content sharing c) A network of computers connected via the internet d) A protocol for secure communication online	2	DS4145 .1
Q 2	How can attributes of nodes influence network analysis?	2	DS4145 .2
Q 3	_____ networks focus on a single node and its direct connections.	2	DS4145 .1
Q 4	What is meant by network density?	2	DS4145 .1
Q 5	Link prediction aims to: a) Add random edges to a network b) Forecast future connections in a network c) Identify isolated nodes d) Optimize graph layouts	2	DS4145 .2
SECTION B (Concept Based Questions)			
Q 6	Given a network of 4 nodes: A, B, C, and D, with the following edges: • A→B; B→C; C→D; D→A 1. Calculate the PageRank scores after one iteration with d=0.8 2. Calculate the PageRank scores after one iteration with d=0.9. 3. Compare the results: How does changing the damping factor affect the PageRank scores?	8	DS4145 .2
Q 7	Hub scores: h(A)≈0.577, h(B)≈0.577, h(C)≈0.577, h(D)=0. Authority scores: a(A)=0, a(B)≈0.408, a(C)≈0.408, a(D)≈0.816. Which node is the best hub? Which node is the best authority?	6	DS4145 .2
Q 8	For a small network, node P has: • Degree = 3 (out of 6 total nodes). • Sum of shortest path distances to all other nodes = 10.	8	DS4145 .1

	Lies on 4 out of 12 shortest paths. Calculate degree centrality, closeness centrality, and betweenness centrality for node P.		
Q 9	Explain how homophily and influence contribute to the evolution of networks over time. How do these processes reinforce or alter network structures, and what are their long-term consequences for social and organizational dynamics?	8	DS4145 .4
SECTION-C (Analytical Based Questions)			
Q 10	For the following Network: - Node A is connected to nodes B and C. - Node B is connected to node D. - Node C is connected to nodes E and F. - Node D is connected to node G. Calculate the SimRank between nodes A and G, considering multiple indirect paths, with a decay factor $C=0.8$	10+10	DS4145 .3
Q 11	What do you mean by anomaly detection in the context of social network? State any four differences between static and dynamic graphs in the context of anomaly detection?		DS4145 .4
SECTION-D (Case Study/Application based Question)			
Q 12	- Explain the recursive formula of SimRank. How does it measure the similarity between two nodes in a graph? - Compare SimRank with other similarity measures like Jaccard Index and Cosine Similarity. In what scenarios is SimRank more suitable? - What role does the damping factor play in the PageRank algorithm? How does its choice affect the results?	7+7+6	DS4145 .4